Mahmoud Soliman

Backend Developer .NET | ASP.NET Core | Clean Architecture | AWS Certified Cloud Practitioner

01010530107 · mahmoudsoliman.dev@gmail.com · https://linkedin.com/in/mahmoud-soliman-backend· https://github.com/MAHMOUD-AHMEDD

SUMMARY

As a backend developer who graduated in mid-2026 with a degree in Computer Science & Artificial Intelligence, I don't just write code; I want to understand exactly where and how it lives in production. I dive deep into .NET to build strict, multi-layer Clean Architecture, then leverage my AWS cloud knowledge — backed by my Cloud Practitioner certification — to deploy and scale those systems across real infrastructure. I am driven by a desire to see the "whole picture" — from the first line of C# to the cloud — and I am looking for a team where I can continuously learn and tackle complex architectural challenges.

EDUCATION

B.Sc. Computer Science & Artificial Intelligence — Fayoum University *2022 – 2026*

TRAINING & EXPERIENCE

AWS Cloud Trainee — NTI (National Telecommunication Institute) *Dec 2025 – Feb 2026*

- Completed a 120-hour AWS program through the Digital Egypt Youth Program, scoring 98% and earning the AWS Certified Cloud Practitioner certification.
- Gained hands-on knowledge of EC2, VPC, IAM, S3, RDS, ECS, Lambda, and cloud architecture principles including high availability, fault tolerance, and cost optimization.

Backend Developer Trainee — DEPI (Digital Egypt Pioneers Initiative) *Apr 2024 – Oct 2024*

- Completed an intensive 3-month government-backed backend development program focused on PHP and Laravel.
- Built and deployed real-world applications including an e-commerce platform and a university management system.

PROJECTS

Smart Career Path — AI-Powered Career Guidance Platform

.NET 8 · ASP.NET Core · Clean Architecture · EF Core · SQL Server · SignalR · JWT · Docker · AWS EC2 + RDS

- Architected for Production: Built a strict 4-layer Clean Architecture backend using ASP.NET Core Identity with TPT inheritance, separating Seekers, Mentors, and Admins into typed entities, secured with role-based JWT auth, refresh token rotation, and a global exception middleware returning consistent error shapes.
- Engineered Complex Features End-to-End: Implemented real-time chat using SignalR with DB-persisted message history, a mentor request-and-approval workflow with a DB-stored notification system, AI-powered career track recommendations via a secured API key endpoint, and MailKit email confirmation and 6-digit password reset flows.
- Deployed to the Cloud: Containerized the full system with Docker using a multi-stage build, deployed to AWS EC2 with SQL Server on AWS RDS Free Tier, and coordinated API integration with a 7-person team (frontend React + AI service) through a shared Swagger contract and SignalR hub documentation.

  GitHub: https://github.com/MAHMOUD-AHMEDD/SmartCareerPath

TaskFlow — Project Management REST API

.NET · ASP.NET Core · EF Core · Clean Architecture · Docker · AWS ECS Fargate · SQL Server

- Architected for Maintainability: Built a multi-tenant backend from scratch using a strict 4-layer Clean Architecture, integrating Unit of Work and Generic Repository patterns to ensure an enterprise-grade codebase.
- Engineered Core Features: Designed a granular, 4-tier role-based access control (RBAC) system using ASP.NET Core Identity, and expanded regional accessibility with dual Arabic/English localization.
- Designed for the Cloud: Containerized the system with Docker and mapped out a highly available production deployment architecture across AWS (ECS Fargate, ALB, and RDS).

  GitHub: https://github.com/MAHMOUD-AHMEDD/TaskManagement

University Learning Platform (Bilingual REST API)

PHP · Laravel · MySQL · REST API · Multi-role auth

- Built a bilingual (Arabic/English) API-based platform connecting doctors, students, and admins with role-based access control.
- Implemented real-time notifications when students subscribe to subjects, with full user and content management.
- Github: https://github.com/MAHMOUD-AHMEDD/University_Learning

E-Commerce Platform

PHP · Laravel · MySQL · MVC

- Developed a fully functional multi-role e-commerce platform (client, supplier, admin) with product management, shopping cart, order processing, and admin panel.
- Github: https://github.com/MAHMOUD-AHMEDD/e-commerce

TECHNICAL SKILLS

- **Primary stack:** .NET 8, C#, ASP.NET Core, Entity Framework Core, LINQ, SQL Server
- **Architecture:** Clean Architecture, MVC, REST API design, Generic Repository, Unit of Work, CQRS
- **Auth & APIs:** ASP.NET Core Identity, JWT, OAuth2 , Swagger / OpenAPI, FluentValidation, AutoMapper
- **DevOps & Cloud:** Docker, Docker Compose, AWS (EC2, VPC, RDS, ECS Fargate, S3, IAM, CloudWatch, Secrets Manager)
- **Databases:** SQL Server, MySQL, Entity Framework Core migrations
- **Other languages:** Python, PHP, Laravel, C++
- **Tools:** Git, GitHub, Visual Studio, VS Code

ACHIEVEMENTS & COMMUNITY

- ECPC / ACPC Competitive Programming — participated in Egypt & Arab Collegiate Programming Contests; developed advanced problem-solving and algorithmic skills in C++.
- ECPC Community Volunteer (Sep 2023 – Sep 2024) — supported community events and competitive programming outreach.
- AWS Certified Cloud Practitioner — earned after completing a 120-hour program at NTI through the Digital Egypt Youth initiative, finishing top of cohort with a 98% score.